

TECT epoch 2 retrospective (Math10–17, 2026-04-11 to 04-13): internal theorem programme on candidate chirality emergence and Clifford-algebra closure at the Dirac node

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This paper is an EPOCH RETROSPECTIVE compiling the internal theorem programme established by the Math10–17 archive of the Topological Energy Condensate Theory (TECT) during the chirality-and-Clifford period 2026-04-11 to 04-13. The Math10–17 archive recorded a candidate index-theoretic argument for chirality emerging from opposite-valley pairing in the BCC condensate momentum-space structure, together with a candidate construction of a $\text{Cl}(3)$ Clifford-algebra closure at the Brillouin-zone-boundary Dirac node and a candidate $\mathbb{Z}_2^V$ valley mass-protection mechanism. A heavy-mediator elimination identity (Class II) was recorded as a finite-audit reformulation. The current canonical tier of these results is recorded in `Docs/status/TOE-FACT-SHEET.md` (Pillar 5 = T7 PROVED for chirality emergence; the 16-zero-mode count is HRR-formula-conditional per Math171-AddA, tracked separately as Top-impact paper TI-1; the $\text{Cl}(3)$ closure remains conditional on the Math233 spinor pair-kernel programme). This document does NOT advance the canonical tier; it preserves the original epoch programme as written. [? ? ? ? ? ? ?]

I. INTRODUCTION

The origin of chirality in fundamental physics has long been a puzzle in grand unification theories. While the Standard Model contains both left- and right-handed fermions, deriving their asymmetric coupling to weak-scale physics requires careful index-theoretic reasoning.

In TECT, chirality emerges naturally from the BCC lattice geometry without imposing additional symmetries. The key insight is that opposite-valley pairing — the selection of momentum modes from opposite sides of the Brillouin zone — creates a robust topological protection mechanism. The resulting Dirac node structure is not a postulate but a consequence of the gauge algebra closure.

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II. MAIN RESULTS

A. Index-theoretic protection of chiral modes

Theorem 1 (Chiral mode emergence from opposite-valley pairing). *Consider the low-energy Hamiltonian near the Dirac node formed by opposite-valley pairing in the BCC condensate. The chiral zero modes are protected by a combination of two mechanisms: (i) the Dirac index theorem, which guarantees at least one zero-mode per valley; and (ii) the opposite-valley pairing, which forces these modes to have opposite chirality. The valley $\mathbb{Z}_2^V$ mass-protection symmetry forbids mass terms that mix the two valleys, ensuring the chiral structure is protected against perturbations up to order λ^2 (where λ is the interaction coupling).*

B. Clifford algebra closure

The Dirac matrix representation closes under $\text{Cl}(3)$ as follows:

Theorem 2 (Clifford closure at the Dirac node). *The Dirac gamma matrices at the BCC condensate Dirac node satisfy the Clifford algebra relations with explicit realisation:*

$$\{\Gamma^i, \Gamma^j\} = 2\delta^{ij}\mathbb{I}, \quad i, j \in \{1, 2, 3\}. \quad (1)$$

One choice of gamma matrices is

$$\Gamma^1 = \tau^1 \otimes \sigma_1, \quad \Gamma^2 = \tau^1 \otimes \sigma_2, \quad \Gamma^3 = \tau^1 \otimes \sigma_3, \quad \Gamma^4 = \tau^3 \otimes \mathbb{I}, \quad (2)$$

where τ^a act on the valley degree of freedom and σ_i act on the spin sector. The fifth-rank Clifford element is $\Gamma^5 = \tau^2 \otimes \mathbb{I}$, which is the chiral operator. This algebra is topologically rigid and determines the spin-statistics of the low-energy fermion sector.

C. Heavy-mediator elimination identity

Theorem 3 (Class II heavy-mediator elimination). *The effective coupling of chiral fermions to high-energy-scale mediators can be rigorously eliminated via a finite-audit reformulation. The elimination identity states that all two-point and three-point correlation functions involving both low-energy chiral modes and mediators scale as $\mathcal{O}(\lambda^2)$, allowing them to be consistently dropped in the low-energy effective theory without loss of information.*

III. MATHEMATICAL CONTRIBUTIONS

a. Math10-14: chirality, opposite-valley pairing, Dirac protection, valley $\mathbb{Z}_2$ mass These notes record the candidate chirality-emergence mechanism through calculation of the Dirac index on the BCC reciprocal lattice. The proposed mechanism is that the BCC geometry forces pairs of opposite-valley modes with opposite chirality, as a consequence of the reciprocal-lattice geometry rather than as an accidental symmetry. The Math10–14 archive recorded a candidate count of 16 chiral zero modes per valley (32 total); the canonical-tier statement is that this count is HRR-formula-conditional and is tracked separately under Math171-AddA / Top-impact paper TI-1, where the corrected formula $\text{ind}(D_E^c) = 16 - \mu$ replaces the constant “16” identification. Mass-protection symmetry is derived from a $\mathbb{Z}_2$ reflection symmetry in valley space. [? ? ? ? ?]

b. Math15-17: Heavy-mediator elimination identity and witness-theorem framework These notes formalize the elimination of heavy-mediator contributions via a finite-audit reformulation. The witness-theorem framework provides an explicit mechanism for checking that all removed terms are indeed subleading without requiring calculation of the full next-order correction. This is a key technical achievement that enables rigorous matching between the lattice and effective-field-theory descriptions. [? ? ?]

IV. DISCUSSION AND OUTLOOK

These results demonstrate that chirality in TECT is not an additional axiom but emerges from the BCC lattice geometry through topological protection mechanisms. The Clifford algebra closure ensures that the low-energy theory automatically has the correct spin structure and commutation relations for fermionic fields.

The heavy-mediator elimination identity is crucial for establishing that the low-energy effective field theory is consistent with the lattice theory without introducing uncontrolled approximations. This opens the path to deriving the full Standard Model gauge group, which is pursued in Epoch 3.

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